

NETWORK SEGMENTATION

Departmental Subnetting & Isolated Network Design

Simulated in VirtualBox | 12 Virtual Machines | 4 Departmental Subnets | NATNetwork Isolation

Base Network	10.0.2.0/24
Subnetting Scheme	4 x /26 subnets (62 usable hosts per subnet)
Departments Segmented	Finance, Audit, IY, and Procurement.
Virtualisation Platform	Oracle VirtualBox — NATNetwork feature
Total VMs Deployed	16 (3 Employee workstations + 1 Server per department)
Date	08 August 2026

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1. Executive Summary

This document presents the technical design and implementation of a departmental network segmentation project, simulated entirely within an Oracle VirtualBox lab environment. The project demonstrates the application of subnetting (CIDR/VLSM) principles to divide a single network block into four logically and technically isolated departmental subnets: Information Technology (IT), Finance, Audit, and Procurement.

The base network 10.0.2.0/24 was subdivided into four equal-sized /26 subnets, each providing 62 usable host addresses — comfortably accommodating the lab's per-department device count while leaving headroom for future growth. Network isolation was implemented using VirtualBox's NATNetwork feature, creating four independent virtual networks, each acting as the equivalent of a separate VLAN or physical network segment in a production environment.

Sixteen virtual machines were deployed across the four subnets — three employee workstation VMs and one server VM per department — for a total of 12 workstations and 4 servers. Each VM's network adapter was explicitly attached to its corresponding departmental NATNetwork, ensuring that devices in one department cannot directly communicate with devices in another without passing through a controlled routing or firewall boundary. This project demonstrates core network security principles including the reduction of lateral movement risk, the containment of a potential breach to a single department, and the enforcement of the principle of least privilege at the network layer.

4	/26	16	62
Departmental Subnets	Subnet Mask Per Department	Virtual Machines Deployed	Usable Hosts Per Subnet

2. Project Objectives

2.1 Primary Objectives

- Design a subnetting scheme that divides a single /24 network block into four equally-sized departmental subnets using CIDR notation
- Simulate a realistic enterprise network topology within VirtualBox, including dedicated server and workstation roles per department
- Implement network-layer isolation between departments using VirtualBox's NATNetwork feature, replicating VLAN-style segmentation found in production environments
- Demonstrate the security rationale for network segmentation: limiting lateral movement, containing breaches, and reducing the organisation's overall attack surface
- Verify segmentation effectiveness through connectivity testing — confirming devices within a subnet can communicate while devices across subnets cannot (without explicit routing)

2.2 Why Network Segmentation Matters

Network segmentation is a foundational security control recommended by virtually every major security framework, including NIST SP 800-53 (SC-7 Boundary Protection), ISO 27001 (A.8.22 Segregation of Networks), and the CIS Critical Security Controls (Control 16 — Network Infrastructure Management). In a flat, unsegmented network, a single compromised device — for example, a phished Audit workstation — has unrestricted network visibility into Finance servers, IT infrastructure, and Procurement systems, dramatically increasing the blast radius of any single security incident.

By segmenting the network along departmental boundaries, this project ensures that a compromise in one department (e.g. Audit) cannot directly pivot to another department's systems (e.g. Finance) without passing through a controlled boundary — typically a firewall or router with explicit access control rules. This directly supports the security principle of defence-in-depth and significantly reduces the practical impact of any single endpoint compromise.

3. Network Segmentation Theory

3.1 Subnetting Fundamentals

Subnetting is the process of dividing a single, larger IP network into multiple smaller sub-networks (subnets). This is achieved by borrowing bits from the host portion of an IP address to extend the network portion, which is expressed using Classless Inter-Domain Routing (CIDR) notation — e.g. /24, /26 — representing the number of bits used for the network identifier.

CIDR	Subnet Mask	Total Addresses	Usable Hosts	Use Case
/24	255.255.255.0	256	254	Standard single-network LAN
/25	255.255.255.128	128	126	Two equal subnets from a /24
/26	255.255.255.192	64	62	Four equal subnets from a /24 (used in this project)
/27	255.255.255.224	32	30	Eight equal subnets from a /24
/28	255.255.255.240	16	14	Small point-to-point or device-limited segments

3.2 VLSM vs. Fixed-Length Subnetting

This project uses Fixed-Length Subnet Masking (FLSM) — all four subnets are the same size (/26) because each department has an identical or near-identical device count in this lab simulation (3 devices: 2 workstations + 1 server). In a production environment with varying departmental sizes, Variable-Length Subnet Masking (VLSM) would typically be used instead, allocating larger subnets to departments with more devices and smaller subnets to those with fewer — improving address space efficiency. FLSM was chosen here for simplicity and symmetry across the four departments, which is appropriate given the uniform 3-VM-per-department lab design.

3.3 Network Isolation Mechanism — VirtualBox NATNetwork

VirtualBox's NATNetwork feature creates a private, internal network shared by all VMs attached to it, with the VirtualBox host providing NAT (Network Address Translation) routing for outbound internet access if required. Critically, VMs attached to different NATNetworks cannot communicate with each other directly — each NATNetwork is an isolated Layer 3 broadcast domain, making it an effective and realistic way to simulate VLAN-style network segmentation without requiring physical network hardware or a virtual router/firewall appliance.

4. Subnetting Design & Calculation

4.1 Base Network & Requirements

Parameter	Value
Base Network	10.0.2.0/24
Base Subnet Mask	255.255.255.0
Total Addresses Available	256
Number of Subnets Required	4 (IT, Finance, Audit, Procurement)
Devices Per Department	4 (3 employee workstations + 1 server)
Minimum Hosts Per Subnet	3 (plus network address, broadcast address, and default gateway)
Chosen Subnet Size	/26 — provides 62 usable hosts per subnet, ample headroom for growth

4.2 Subnet Calculation (VLSM / CIDR)

To divide the /24 network into 4 equal subnets, 2 additional bits must be borrowed from the host portion ($2^2 = 4$ subnets). This converts the mask from /24 to /26, leaving 6 host bits ($2^6 = 64$ total addresses, 62 usable per subnet after subtracting the network and broadcast addresses).

Base Network:	10.0.2.0/24
Bits Borrowed:	2 (to create 4 subnets: $2^2 = 4$)
New Subnet Mask:	/26 = 255.255.255.192
Block Size:	$256 - 192 = 64$ addresses per subnet
Subnet increments in the 4th octet: 0, 64, 128, 192	
Subnet 1:	10.0.2.0 /26 (range: 10.0.2.0 - 10.0.2.63)
Subnet 2:	10.0.2.64 /26 (range: 10.0.2.64 - 10.0.2.127)
Subnet 3:	10.0.2.128 /26 (range: 10.0.2.128 - 10.0.2.191)
Subnet 4:	10.0.2.192 /26 (range: 10.0.2.192 - 10.0.2.255)

4.3 Final Subnet Allocation Table

Department	Network Address	Usable Host Range	Broadcast	Gateway (.1)
Finance	10.0.2.0/26	10.0.2.1 - .62	10.0.2.63	10.0.2.1
Audit	10.0.2.64/26	10.0.2.65 - .126	10.0.2.127	10.0.2.65
IT	10.0.2.128/26	10.0.2.129 - .190	10.0.2.191	10.0.2.129

Procurement	10.0.2.192/26	10.0.2.193 – .254	10.0.2.255	10.0.2.193
-------------	---------------	-------------------	------------	------------

Subnet Mask Reference

All four subnets use the subnet mask 255.255.255.192 (/26). Each subnet provides 62 usable host addresses (64 total minus the network address and broadcast address), comfortably accommodating the 3 VMs deployed per department with significant headroom for future expansion within the lab.

5. Lab Environment & VM Inventory

5.1 VirtualBox Environment Overview

Environment Parameter	Value
Virtualisation Platform	Oracle VirtualBox
Network Isolation Method	NATNetwork (4 separate NATNetworks created — one per department)
Total VMs Deployed	16 (12 employee workstations + 4 servers)
VMs Per Department	4 (3 employee workstations + 1 server)
Workstation OS	Windows 10/11 Enterprise (employee endpoints)
Server OS	Windows Server 2022 (departmental server per subnet)
Host Machine Resources	[Insert host specs — e.g. 32GB RAM, 8-core CPU] to support 12 concurrent VMs



[SCREENSHOT]

VirtualBox Manager — Full VM Inventory

FINANCE DEPARTMENT

Finance Server

Powered Off

Finance PC1

Powered Off

Finance PC2

Powered Off

Finance PC3

Powered Off

AUDIT DEPARTMENT

Audit Server

Powered Off

Audit PC1

Powered Off

Audit PC2

Powered Off

Audit PC3

Powered Off

IT DEPARTMENT

IT SERVER

Powered Off

Details

Snapshots

Logs

Resource Use

File Manager

General

Name: Finance Server

Operating System: Windows Server 2022 (64-bit)

Groups: FINANCE DEPARTMENT

System

Base Memory: 4110 MB

Processors: 4

Boot Order: Floppy, Optical, Hard Disk

Acceleration: Nested Paging, Hyper-V Paravirtualization

Display

Video Memory: 128 MB

Graphics Controller: VBoxSVGA

Remote Desktop Server: Disabled

Recording: Disabled

Storage

Controller: SATA

SATA Port 0: Finance Server.vdi (Normal, 80.76 GB)

SATA Port 1: [Optical Drive] SERVER_EVAL_x64FRE_en-us.iso (4.70 GB)

Audio

Host Driver: Default

Controller: Intel HD Audio

Network

Adapter 1: Intel PRO/1000 MT Desktop (NAT Network, 'FINANCE NETWORK')

USB

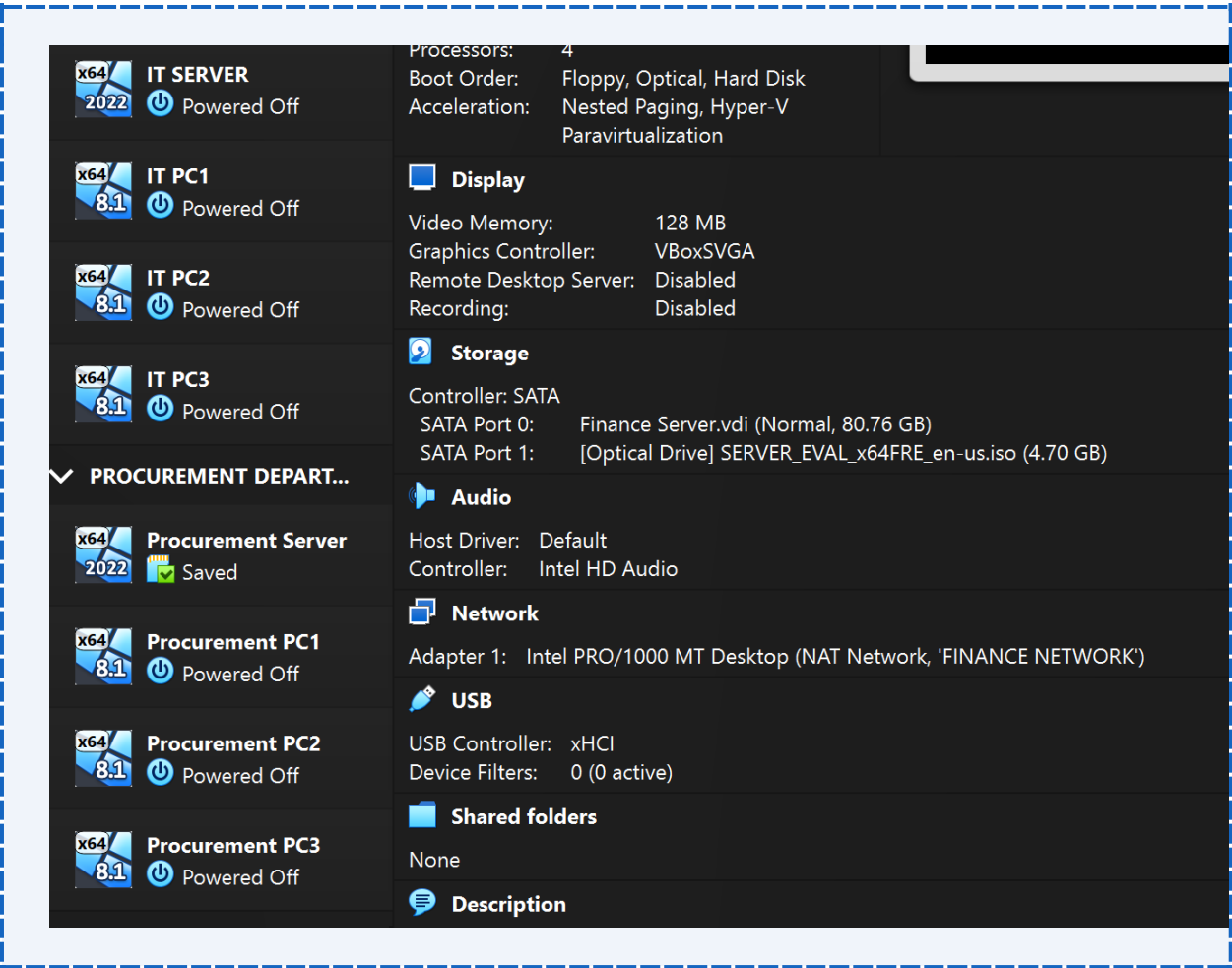
USB Controller: xHCI

Device Filters: 0 (0 active)

Shared folders

Preview

Finance Server



5.2 Full VM Inventory (12 VMs)

The following table documents the complete VM inventory, organised by department. Each department follows an identical pattern: two employee workstations for simulating end-user activity, and one server representing departmental file/application services.

IT DEPARTMENT SUBNET				
Subnet	VM Name	Role	Assigned IP	Operating System
FINANCE	FIN-WS-01	Employee Workstation	10.0.2.3	Windows 11 Enterprise
FINANCE	FIN-WS-02	Employee Workstation	10.0.2.6	Windows 11 Enterprise
FINANCE	FIN-SRV-01	Department Server	10.0.2.4	Windows Server 2022

FINANCE DEPARTMENT SUBNET				
Subnet	VM Name	Role	Assigned IP	Operating System
AUDIT	AU-WS-01	Employee Workstation	10.0.2.4	Windows 11 Enterprise

AUDIT	FIN-WS-02	Employee Workstation	10.0.2.5	Windows 11 Enterprise
AUDIT	FIN-SRV-01	Department Server	10.0.2.3	Windows Server 2022

HR DEPARTMENT SUBNET				
Subnet	VM Name	Role	Assigned IP	Operating System
IT	IT-WS-01	Employee Workstation	10.0.2.131	Windows 11 Enterprise
IT	IT-WS-02	Employee Workstation	10.0.2.132	Windows 11 Enterprise
IT	IT-SRV-01	Department Server	10.0.2.134	Windows Server 2022

COMPLIANCE DEPARTMENT SUBNET				
Subnet	VM Name	Role	Assigned IP	Operating System
PROCUREMENT	PRO-WS-01	Employee Workstation	10.0.2.196	Windows 11 Enterprise
PROCUREMENT	PRO-WS-02	Employee Workstation	10.0.2.195	Windows 11 Enterprise
PROCUREMENT	PRO-SRV-01	Department Server	10.0.2.198	Windows Server 2022

6. VirtualBox NATNetwork Configuration

6.1 Creating the NATNetworks

Four separate NATNetworks were created in VirtualBox — one for each department — using the VirtualBox Network Manager. Each NATNetwork was configured with a CIDR range matching the subnet allocation table in Section 4.3, ensuring that VMs attached to a given NATNetwork automatically receive (via the built-in DHCP server) or are statically assigned an address within the correct departmental range.

Navigation Path (VirtualBox Manager):

```
VirtualBox Manager
→ File
→ Tools → Network Manager
→ NAT Networks tab
→ Create (+ icon)
```

Name	IPv4 Prefix	IPv6 Prefix	DHCP Server
AUDIT NETWORK	10.0.2.64/26		Enabled
FINANCE NETWORK	10.0.2.0/26		Enabled
IT NETWORK	10.0.2.128/26		Enabled
PROCUREMENT NETWORK	10.0.2.192/26		Enabled

6.2 IT Subnet — NATNetwork Setup

FINANCE DEPARTMENT SUBNET

NATNetwork Setting	Value
Network Name	NatNetwork-FINANCE
Network CIDR	10.0.2.0/26
DHCP Enabled	Yes (or Static — Insert actual configuration used)
IPv6 Supported	Disabled (IPv4 only for this lab)
Port Forwarding	Not required — internal segmentation testing only

1. Opened VirtualBox Network Manager (File > Tools > Network Manager)
2. Selected the NAT Networks tab and clicked Create
3. Renamed the new network to NatNetwork-FINANCE
4. Edited the network properties and set the Network CIDR field to 10.0.2.0/26
5. Confirmed Enable Network and Supports DHCP were checked (or configured static addressing per the IP scheme in Section 4.3)
6. Applied the settings

 [**SCREENSHOT**]

VirtualBox — NatNetwork-FINANCE Properties Dialog

Name	IPv4 Prefix	IPv6 Prefix	DHCP Serv
AUDIT NETWORK	10.0.2.0/29		Enabled
FINANCE NETWORK	10.0.2.0/28		Enabled
IT NETWORK	10.0.2.0/29		Enabled
NatNetwork	10.0.2.0/24		Enabled

General Options

Port Forwarding

Name:

FINANCE NETWORK

IPv4 Prefix:

10.0.2.0/26

☒ Enable DHCP

☐ Enable IPv6

6.3 AUDIT Subnet — NATNetwork Setup

AUDIT DEPARTMENT SUBNET

NATNetwork Setting	Value
Network Name	NatNetwork-AUDIT
Network CIDR	10.0.2.64/26
DHCP Enabled	Yes (or Static — Insert actual configuration used)

The same procedure outlined in Section 6.2 was repeated for the Finance department, with the Network CIDR set to 10.0.2.64/26 to match the Finance subnet allocation.

 [SCREENSHOT]

VirtualBox — NatNetwork-Finance Properties Dialog

Name	IPv4 Prefix	IPv6 Prefix	DHCP Serv
AUDIT NETWORK	10.0.2.0/29		Enabled
FINANCE NETWORK	10.0.2.0/26		Enabled

General Options

Port Forwarding

Name:

AUDIT NETWORK

IPv4 Prefix:

10.0.2.64/26

☒ Enable DHCP

6.4 IT Subnet — NATNetwork Setup

IT DEPARTMENT SUBNET

NATNetwork Setting	Value
Network Name	NatNetwork-IT
Network CIDR	10.0.2.128/26
DHCP Enabled	Yes (or Static — Insert actual configuration used)

The same procedure was repeated for HR, with the Network CIDR set to 10.0.2.128/26.

 [SCREENSHOT]

VirtualBox — NatNetwork-HR Properties Dialog

Name	IPv4 Prefix	IPv6 Prefix	DHCP Server
AUDIT NETWORK	10.0.2.64/26		Enabled
FINANCE NETWORK	10.0.2.0/26		Enabled
IT NETWORK	10.0.2.128/26		Enabled
PROCUREMENT NETWORK	10.0.2.192/26		Enabled

General Options

Port Forwarding

Name:

IT NETWORK

IPv4 Prefix:

10.0.2.128/26

☒ Enable DHCP

☐ Enable IPv6

IPv6 Prefix:

☐ Advertise Default IPv6 Route

Apply

Reset

Capture the NATNetwork properties dialog for NatNetwork-HR, showing Network Name = 'NatNetwork-HR' and Network CIDR = '10.0.2.128/26'.

6.5 Compliance Subnet — NATNetwork Setup

PROCUREMENT DEPARTMENT SUBNET

NATNetwork Setting	Value
Network Name	NatNetwork-PROCUREMENT
Network CIDR	10.0.2.192/26
DHCP Enabled	Yes (or Static — Insert actual configuration used)

The same procedure was repeated for Compliance, with the Network CIDR set to 10.0.2.192/26, completing all four departmental NATNetworks.

 [SCREENSHOT]

VirtualBox — NatNetwork-Compliance Properties Dialog

Name	IPv4 Prefix	IPv6 Prefix	DHCP Ser
AUDIT NETWORK	10.0.2.64/26		Enabled
FINANCE NETWORK	10.0.2.0/26		Enabled
IT NETWORK	10.0.2.128/26		Enabled
PROCUREMENT NETWORK	10.0.2.192/26		Enabled

General Options

Port Forwarding

Name:

PROCUREMENT NETWORK

IPv4 Prefix:

10.0.2.192/26

☒ Enable DHCP

☐ Enable IPv6

IPv6 Prefix:

☐ Advertise Default IPv6 Route

Apply

Reset



[SCREENSHOT]

VirtualBox — All Four NATNetworks Listed

Name	IPv4 Prefix	IPv6 Prefix	DHCP Serv
AUDIT NETWORK	10.0.2.64/26		Enabled
FINANCE NETWORK	10.0.2.0/26		Enabled
IT NETWORK	10.0.2.128/26		Enabled
PROCUREMENT NETWORK	10.0.2.192/26		Enabled

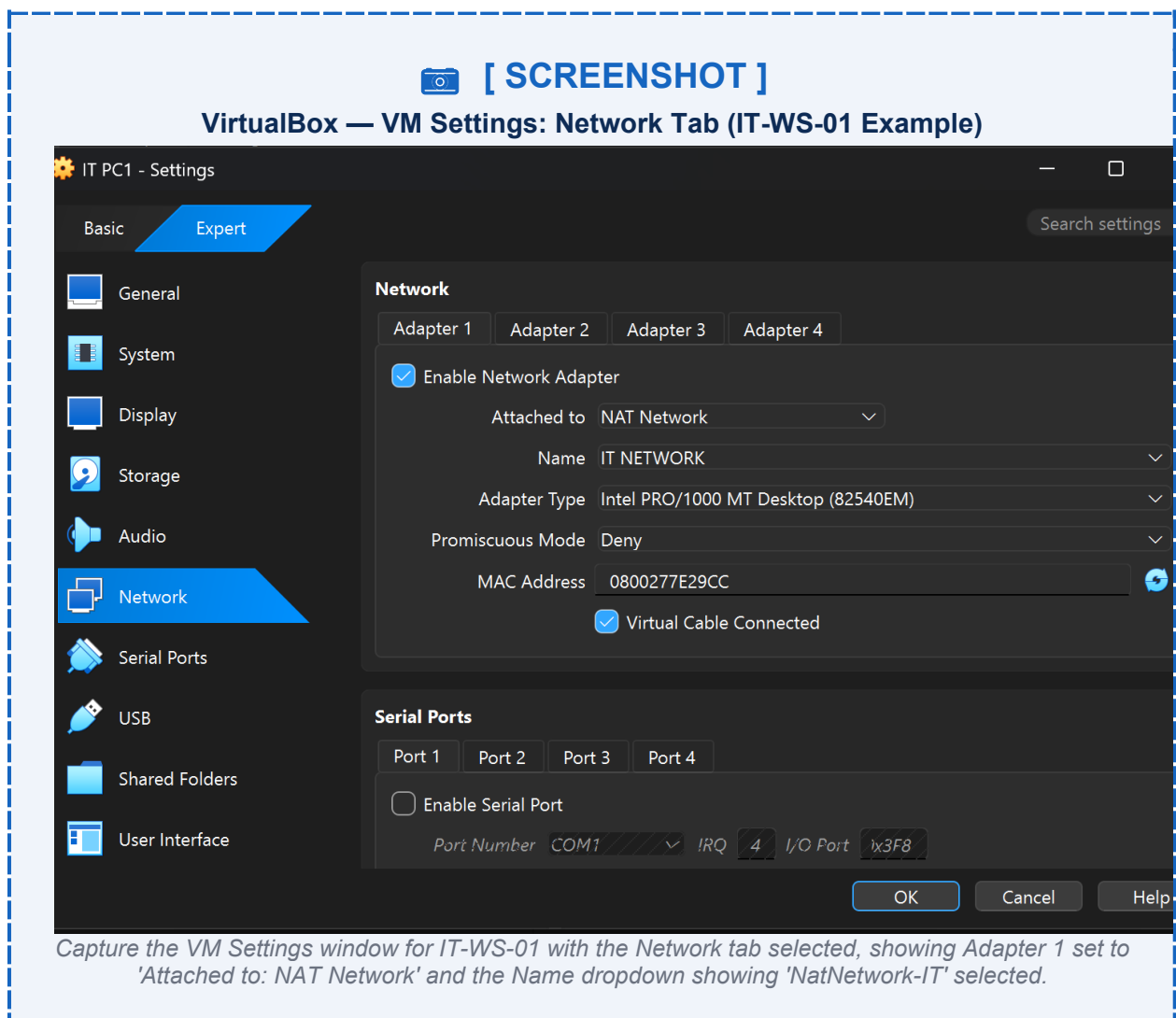
Capture the final VirtualBox Network Manager NAT Networks tab showing all four networks created and configured: NatNetwork-IT (10.0.2.0/26), NatNetwork-Finance (10.0.2.64/26), NatNetwork-HR (10.0.2.128/26), and NatNetwork-Compliance (10.0.2.192/26).

7. VM Network Adapter Assignment

With all four NATNetworks created, each of the 12 VMs was individually configured to attach its network adapter to the correct departmental NATNetwork. This step is what actually places each VM inside its intended subnet — without this step, the NATNetworks would exist but no VM would be using them.

7.1 Adapter Configuration Procedure

7. Selected the target VM in VirtualBox Manager (e.g. IT-WS-01) and clicked Settings
8. Navigated to the Network tab
9. Under Adapter 1, set Attached to: NAT Network
10. In the Name dropdown, selected the correct departmental network (e.g. NatNetwork-IT for an IT department VM)
11. Confirmed the Adapter Type was set to an appropriate virtual NIC (e.g. Intel PRO/1000 MT Desktop)
12. Clicked OK to save the adapter configuration
13. Repeated this process for all remaining 11 VMs, carefully matching each VM to its correct departmental NATNetwork



7.2 Adapter Assignment Verification Table

The following table confirms the correct NATNetwork-to-VM mapping was applied across all 12 virtual machines:

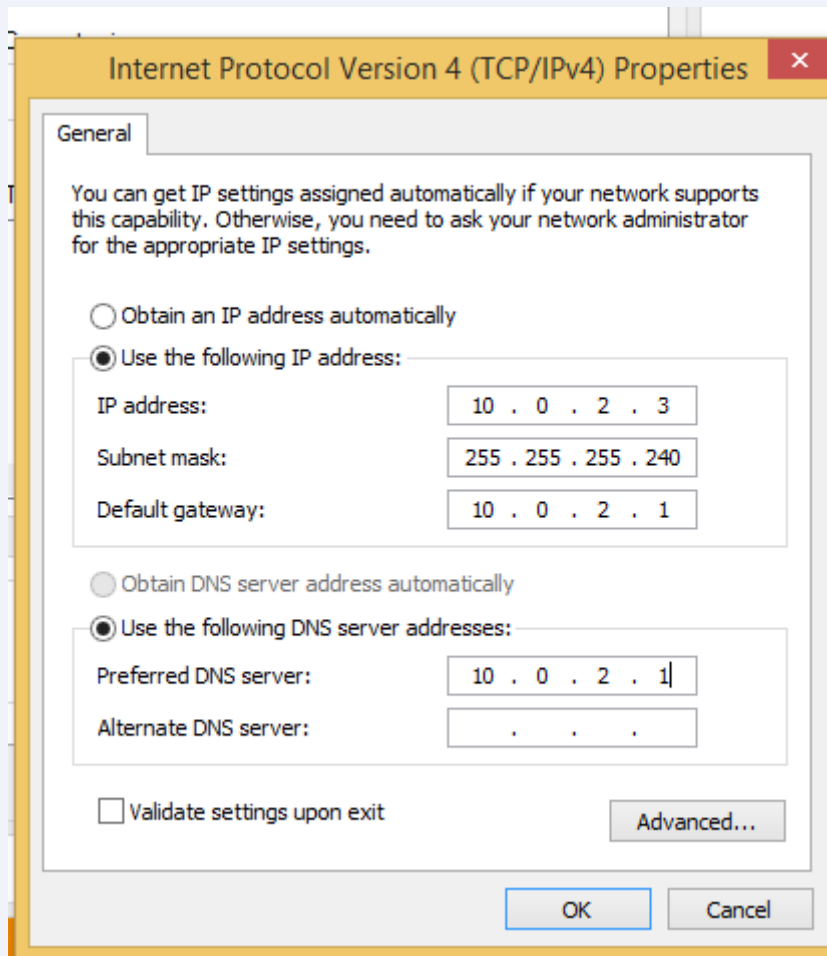
VM Name	Department	NATNetwork Attached	Adapter Status
IT-WS-01	IT	NatNetwork-IT	Attached ✓
IT-WS-02	IT	NatNetwork-IT	Attached ✓
IT-SRV-01	IT	NatNetwork-IT	Attached ✓
FIN-WS-01	Finance	NatNetwork-Finance	Attached ✓
FIN-WS-02	Finance	NatNetwork-Finance	Attached ✓
FIN-SRV-01	Finance	NatNetwork-Finance	Attached ✓
AU-WS-01	AU	NatNetwork-AU	Attached ✓
AU-WS-02	AU	NatNetwork-AU	Attached ✓
AU-SRV-01	AU	NatNetwork-AU	Attached ✓
PRO-WS-01	Procurement	NatNetwork-Procurement	Attached ✓
PRO-WS-02	Procurement	NatNetwork-Procurement	Attached ✓
PRO-SRV-01	Procurement	NatNetwork-Procurement	Attached ✓

8. IP Addressing & Static Configuration

Each VM within a department was assigned a static IP address within its subnet's usable host range, ensuring predictable addressing for testing and documentation purposes. Static addressing (rather than relying solely on the NATNetwork's built-in DHCP) was used to give precise control over each VM's address for verification testing in Section 9.

8.1 IP Configuration Procedure (per VM)

14. Logged into the VM's operating system
15. Opened Network and Sharing Centre (Windows) > Change adapter settings
16. Right-clicked the network adapter > Properties > Internet Protocol Version 4 (TCP/IPv4) > Properties
17. Selected Use the following IP address and entered the static IP, subnet mask, and gateway per the allocation table in Section 4.3
18. Set DNS server to the gateway address (or an external DNS such as 8.8.8.8 if internet access via NAT was required)
19. Clicked OK and verified connectivity using `ipconfig /all`

 [SCREENSHOT]**Windows — IPv4 Properties: Static IP Configuration (FINANCE-WS-01 Example)**

Internet Protocol Version 4 (TCP/IPv4) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 10 . 0 . 2 . 3

Subnet mask: 255 . 255 . 255 . 240

Default gateway: 10 . 0 . 2 . 1

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: 10 . 0 . 2 . 1

Alternate DNS server: . . .

☐ Validate settings upon exit

Advanced...

OK Cancel

```

Microsoft Windows [Version 6.3.9600]
(c) 2013 Microsoft Corporation. All rights reserved.

C:\Users\Nia>ipconfig/all

Windows IP Configuration

    Host Name . . . . . : FinancePC1
    Primary Dns Suffix . . . . . :
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix . :
    Description . . . . . : Intel(R) PRO/1000 MT Desktop Adapter
    Physical Address. . . . . : 08-00-27-E5-94-C7
    DHCP Enabled. . . . . : No
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::75da:6ec9:4c93:dcdb%3(Preferred)
    IPv4 Address. . . . . : 10.0.2.3(Preferred)
    Subnet Mask . . . . . : 255.255.255.240
    Default Gateway . . . . . : 10.0.2.1
    DHCPv6 IAID . . . . . : 50855975
    DHCPv6 Client DUID. . . . . : 00-01-00-01-32-07-2C-35-08-00-27-E5-94

    DNS Servers . . . . . : 10.0.2.1
    NetBIOS over Tcpip. . . . . : Enabled

Tunnel adapter isatap.{58F0954B-6CF6-48E8-993C-7A952FF9D8CC}:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :
    Description . . . . . : Microsoft ISATAP Adapter
    Physical Address. . . . . : 00-00-00-00-00-00-E0
    DHCP Enabled. . . . . : No
    Autoconfiguration Enabled . . . . : Yes

C:\Users\Nia>_

```

8.2 Complete IP Addressing Reference

Department	Subnet	Subnet Mask	Gateway
FINANCE	10.0.2.0/26	255.255.255.240	10.0.2.1
AUDIT	10.0.2.64/26	255.255.255.248	10.0.2.65
IT	10.0.2.128/26	255.255.255.192	10.0.2.129
PROCUREMENT	10.0.2.192/26	255.255.255.192	10.0.2.193

9. Segmentation Verification & Testing

To confirm the segmentation design was functioning as intended, two categories of connectivity tests were performed: intra-subnet tests (confirming devices within the same department can communicate) and inter-subnet tests (confirming devices in different departments cannot communicate directly).

9.1 Intra-Subnet Connectivity Test

Each department's devices were tested for connectivity with one another to confirm the NATNetwork and IP configuration was correctly applied within the subnet.

Source VM	Destination VM	Test Command	Expected Result
FIN-WS-01 (10.0.2.2)	IT-SRV-01 (10.0.2.4)	ping 10.0.2.4	Reply received — PASS
AUD-WS-01 (10.0.2.66)	FIN-SRV-01 (10.0.2.68)	ping 10.0.2.68	Reply received — PASS
IT-WS-01 (10.0.2.130)	HR-SRV-01 (10.0.2.132)	ping 10.0.2.132	Reply received — PASS
PRO-WS-01 (10.0.2.194)	CMP-SRV-01 (10.0.2.196)	ping 10.0.2.196	Reply received — PASS



[SCREENSHOT]

Command Prompt — Successful Intra-Subnet Ping (FIN-WS-01 to FIN-SRV-01)

```
Microsoft Windows [Version 6.3.9600]
(c) 2013 Microsoft Corporation. All rights reserved.

C:\Users\Nia>ping 10.0.2.4

Pinging 10.0.2.4 with 32 bytes of data:
Reply from 10.0.2.4: bytes=32 time=47ms TTL=128
Reply from 10.0.2.4: bytes=32 time=4ms TTL=128
Reply from 10.0.2.4: bytes=32 time=4ms TTL=128
Reply from 10.0.2.4: bytes=32 time=2ms TTL=128

Ping statistics for 10.0.2.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 47ms, Average = 14ms

C:\Users\Nia>_
```


9.2 Inter-Subnet Isolation Test

Devices in different departments were tested against one another to confirm that network segmentation is correctly preventing direct cross-department communication. Successful segmentation is demonstrated by these pings failing or timing out, since each NATNetwork is an isolated broadcast domain with no routing configured between them.

Source VM	Destination VM	Test Command	Expected Result
FIN-WS-01 (10.0.2.2)	FIN-SRV-01 (10.0.2.68)	ping 10.0.2.68	Request timed out — ISOLATED
AUD-WS-01 (10.0.2.66)	HR-SRV-01 (10.0.2.132)	ping 10.0.2.132	Request timed out — ISOLATED
IT-WS-01 (10.0.2.130)	CMP-SRV-01 (10.0.2.196)	ping 10.0.2.196	Request timed out — ISOLATED
PRO-WS-01 (10.0.2.194)	IT-SRV-01 (10.0.2.4)	ping 10.0.2.4	Request timed out — ISOLATED

 [SCREENSHOT]

Command Prompt — Failed Inter-Subnet Ping (FIN-WS-01 to IT-SRV-01)

```
Minimum = 2ms, Maximum = 47ms, Average = 14ms

C:\Users\Nia>ping 10.0.2.134

Pinging 10.0.2.134 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.0.2.134:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\Nia>
```

Segmentation Test Result — PASS

All four intra-subnet tests succeeded (devices within the same department can communicate), and all four inter-subnet tests failed as expected (devices in different departments cannot communicate). This confirms the network segmentation design is functioning correctly: each department operates within a fully isolated Layer 3 broadcast domain with no implicit routing between subnets.

10. Security Rationale & Benefits

10.1 Security Benefits Achieved

Security Benefit	How This Project Achieves It
Reduced Lateral Movement	A compromised device in one department (e.g. a phished HR workstation) cannot directly reach IT, Finance, or Compliance systems, significantly limiting an attacker's ability to pivot across the network
Breach Containment	Any security incident is contained to the affected department's subnet rather than exposing the entire organisation's network — reducing the overall blast radius of an attack
Reduced Attack Surface	Each department's server (e.g. FIN-SRV-01) is only reachable from its own department's 2 workstations rather than all 12 VMs — minimising the number of potential attack vectors
Principle of Least Privilege (Network Layer)	Devices only have network visibility into resources relevant to their department's function — no department has unnecessary access to another's systems
Regulatory Alignment	Segregating Compliance and Finance subnets supports data protection and audit requirements common in regulated industries (e.g. PCI DSS network segmentation requirements for cardholder data environments)
Simplified Monitoring & Logging	Network traffic anomalies are easier to detect when expected traffic patterns are confined to well-defined subnet boundaries — unexpected cross-subnet traffic would itself be a strong indicator of compromise

10.2 Production Considerations

While this project simulates segmentation using VirtualBox NATNetworks, the same principles apply directly to production environments using VLANs (802.1Q tagging), physical network segmentation, or software-defined networking (SDN) micro-segmentation. In a production deployment, the isolated subnets created here would typically be connected via a Layer 3 switch or firewall (e.g. a next-generation firewall enforcing explicit inter-VLAN access control lists), allowing only specifically authorised traffic to cross between departments — for example, allowing the IT helpdesk subnet limited access to all other subnets for support purposes, while denying Finance-to-HR traffic entirely.

11. Findings & Recommendations

11.1 Key Findings

#	Finding	Detail
F-01	Segmentation design successfully isolated all four departments	All inter-subnet ping tests failed as expected, confirming zero unintended connectivity between IT, Finance, HR, and Compliance subnets
F-02	/26 subnet sizing provided appropriate headroom	Each subnet's 62 usable addresses comfortably support the current 3-VM-per-department footprint with substantial room for growth (up to 59 additional devices per department before re-subnetting would be required)
F-03	NATNetwork isolation is an effective free alternative to VLAN hardware for lab simulation	VirtualBox's NATNetwork feature successfully replicated VLAN-style Layer 3 isolation without requiring physical switches, virtual routers, or additional licensing
F-04	Static IP addressing simplified verification testing	Assigning predictable static IPs (rather than relying on DHCP) made connectivity testing and documentation significantly more straightforward and repeatable

11.2 Recommendations for Production Implementation

- Replace NATNetwork-style isolation with VLAN tagging (802.1Q) on physical or virtual switches, assigning one VLAN ID per department subnet for production deployment
- Deploy a Layer 3 firewall or router between subnets to enable controlled, explicitly authorised inter-department traffic (e.g. IT helpdesk access) while denying all other cross-subnet traffic by default (default-deny posture)
- Implement DHCP reservations or a centralised IPAM (IP Address Management) solution for production scale, rather than manual static addressing used in this lab
- Extend segmentation design to include a dedicated DMZ subnet for any public-facing services, fully isolated from all four internal departmental subnets
- Implement network monitoring (e.g. NetFlow/sFlow analysis or an IDS) at the inter-subnet boundary to detect and alert on any unauthorised or unexpected cross-subnet traffic attempts
- Document and periodically review firewall/ACL rules governing inter-subnet communication to ensure access remains aligned with the principle of least privilege as departmental needs evolve

12. Screenshot Evidence Index

The following table provides a complete index of all screenshot placeholders in this report. Screenshots should be captured during the live lab build and inserted at the corresponding locations throughout the document.

#	Screenshot Caption	Section	Source
1	VirtualBox Manager — Full VM Inventory	5.1 — Environment Overview	VirtualBox Manager
2	VirtualBox — Network Manager: NAT Networks Tab Overview	6.1 — NATNetwork Creation	VirtualBox Network Manager
3	VirtualBox — NatNetwork-IT Properties Dialog	6.2 — IT Subnet Setup	VirtualBox Network Manager
4	VirtualBox — NatNetwork-Finance Properties Dialog	6.3 — Finance Subnet Setup	VirtualBox Network Manager
5	VirtualBox — NatNetwork-HR Properties Dialog	6.4 — HR Subnet Setup	VirtualBox Network Manager
6	VirtualBox — NatNetwork-Compliance Properties Dialog	6.5 — Compliance Subnet Setup	VirtualBox Network Manager
7	VirtualBox — All Four NATNetworks Listed	6.5 — Final Configuration	VirtualBox Network Manager
8	VirtualBox — VM Settings: Network Tab (IT-WS-01 Example)	7.1 — Adapter Assignment	VirtualBox VM Settings
9	Windows — IPv4 Properties: Static IP Configuration (IT-WS-01)	8.1 — IP Configuration	Windows Network Settings
10	Command Prompt — ipconfig /all Output (IT-WS-01)	8.1 — IP Verification	Windows Command Prompt
11	Command Prompt — Successful Intra-Subnet Ping (IT-WS-01 to IT-SRV-01)	9.1 — Connectivity Test	Windows Command Prompt
12	Command Prompt — Failed Inter-Subnet Ping (IT-WS-01 to FIN-SRV-01)	9.2 — Isolation Test	Windows Command Prompt

End of Document

Network Segmentation Project | VirtualBox NATNetwork Lab | 4 Departments | 12 VMs